Report on AI Agentic customer support

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Project Title:

Exploratory Data Analysis of Agentic AI Customer Support Ticket Data for Understanding Service Performance and Customer Sentiment

Project Domain:

Artificial Intelligence-based Customer Support Analytic

Submission Date: 30-03-2026
Mentor Name: Kumaran M
Link for Raw Data Set:
https://drive.google.com/file/d/1DapFdX4R3rmArbNdt34gP9qFQtxfsHA9/view?usp=sharing

Link for Cleaned Data Set:
https://drive.google.com/file/d/1lIuN6vLUa_3PfkYGfiEFcDJVBr6uOpB7/view?usp=sharing

Project Title

Exploratory Data Analysis of Agentic AI Customer Support Ticket Data for Understanding Service Performance and Customer Sentiment

Domain

Artificial Intelligence-based Customer Support Analytic

Objective

1. To perform exploratory data analysis (EDA) in order to identify meaningful patterns, trends, and relationships within the Agentic AI customer support dataset.

2. To clean, preprocess, and transform the dataset to ensure data quality and prepare it for accurate analysis.

3. To create informative visualizations that effectively communicate key insights and trends present in the data.

4. To derive actionable insights and recommendations that can help improve customer support performance and service efficiency.

Outcome

- The outcome of this project is a comprehensive understanding of the

- Agentic AI customer support dataset through exploratory data

analysis. The analysis helps identify key patterns, trends, and relationships within the data related to customer queries, service performance, and resolution efficiency.

- Through proper data cleaning, preprocessing, and visualization, the project provides meaningful insights into customer support operations and highlights important factors influencing customer satisfaction.

- The results of this analysis can support better decision-making and help improve the effectiveness of AI-driven customer support systems.

Dataset Information

Source: Kaggle

Year / Timeline: 2025-2026

Dataset Description:

Ticket_ID – It represents the unique identification number assigned to each customer support ticket.

Customer_ID – It indicates the unique identification number assigned to each customer in the system.

Customer_Name – It tells us the name of the customer who raised the support request.

Customer_Email – It provides the email address of the customer used for communication.

Customer_Country – It indicates the country from which the customer is contacting support.

Customer_Language – It shows the preferred language used by the customer during communication.

Channel – It represents the medium through which the customer contacted support (such as email, chat, or phone).

Query_Category – It identifies the main category of the customer's issue or request.

Query_Subcategory – It provides a more detailed classification of the customer's problem.

Priority_Level – It indicates the urgency level of the support ticket (low, medium, or high).

Sentiment_Score – It represents the emotional tone of the customer's message, indicating whether the sentiment is positive, neutral, or negative.

Resolution_Status – It shows whether the issue has been resolved, pending, or still in progress.

Resolution_Time – It indicates the time taken by the support team or AI system to resolve the customer query.

Customer_Feedback_Rating – It represents the rating given by the customer after the issue is resolved.

Timestamp – It records the date and time when the customer support ticket was created.

Type of Analysis

The project primarily uses Descriptive Analysis to summarize and understand the characteristics of the dataset

through statistical measures and visualizations. Additionally, Diagnostic Analysis is used to identify relationships and underlying reasons behind patterns observed in the data.

Descriptive Analysis with elements of Diagnostic Analysis.

Stage 1 – Problem Definition and Dataset Selection

Customer support systems handle a large number of service requests every day. Understanding customer issues, response efficiency, and service quality is important for improving customer satisfaction.

The main problem addressed in this project is to analyze customer support ticket data generated from an Agentic AI support system and identify meaningful patterns related to customer queries, service response time, sentiment, and support resolution.

The analysis aims to help organizations understand customer behavior, improve support performance, and enhance the efficiency of AI-driven customer support services.

Define the business problem and expected outcome
The expected outcome of this project is to obtain meaningful insights from the customer support dataset using exploratory data analysis techniques. The analysis will help identify

patterns in customer queries, support channels, sentiment trends, and resolution time.

The results of the analysis can help organizations improve service efficiency, better understand customer concerns, and make informed decisions to enhance customer support operations.

Choose dataset and explain its source, size, and features

The dataset selected for this project is an Agentic AI Customer Support Dataset.

Source: Kaggle / Open Data Portal

The dataset contains information about customer support interactions including ticket details, customer information, communication channels, query categories, sentiment scores, priority levels, and resolution status.

Dataset Size and Features

Number of Rows: 1200

Number of Columns: 20

The dataset includes both categorical and numerical variables related to customer support operations.

Some important features in the dataset include:

Ticket ID

Customer ID

Customer Name

Customer Email

Customer Country

Customer Language

Channel

Query Category

Query Subcategory

Sentiment Score

Priority Level

Resolution Status

Resolution Time

Customer Feedback Rating

Timestamp

These features help analyze customer support performance and customer experience.

Import libraries, load dataset

Dataset description (rows, columns, features)

Initial EDA (head, info, describe, shape, null checks, duplicate check)

```
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns
```

```
from google.colab import files
uploaded = files.upload()
```
Agentic_AI_Customer_Support_Dataset_v1.csv(text/csv)

```
import pandas as pd
df =
pd.read_csv('Agentic_AI_Customer_Support_Dataset_v1.csv'
)

df =
pd.read_csv('Agentic_AI_Customer_Support_Dataset_v1.csv'
)
df.head()

Df.shape
(1200, 20)

df.info()
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 1200 entries, 0 to 1199
Data columns (total 20 columns):
 #   Column                  Non-Null Count  Dtype
---  ------                  --------------  -----
```

0	Ticket_ID	1200 non-null	object
1	Customer_ID	1200 non-null	object
2	Customer_Name	1200 non-null	object
3	Customer_Email	1200 non-null	objecct
4	Customer_Country	1200 non-null	object
5	Customer_Language	1200 non-null	object
6	Channel	1200 non-null	object
7	Query_Category	1200 non-null	object
8	Query_Subcategory	1200 non-null	object
9	Customer_Query_Text	1200 non-null	object
10	Intent_Detected	1200 non-null	object
11	Sentiment_Score	1200 non-null	float64
12	Priority_Level	1200 non-null	object
13	AI_Response_Text	1200 non-null	object
14	Human_Escalation_Flag	1200 non-null	bool
15	Resolution_Status	1200 non-null	object
16	Resolution_Time	1200 non-null	int64
17	Customer_Feedback_Rating	1200 non-null	int64
18	Follow_Up_Required	1200 non-null	bool
19	Timestamp	1200 non-null	object

dtypes: bool(2), float64(1), int64(2), object(15)

memory usage: 171.2+ KB

```
df.isnull().sum()
```

	0
Ticket_ID	0
Customer_ID	0
Customer_Name	0
Customer_Email	0
Customer_Country	0
Customer_Language	0
Channel	0
Query_Category	0
Query_Subcategory	0

Customer_Query_Text	0
Intent_Detected	0
Sentiment_Score	0
Priority_Level	**0**
AI_Response_Text	0
Human_Escalation_Flag	0
Resolution_Status	0
Resolution_Time	0
Customer_Feedback Ratnig	0
Follow_Up_Requird	0
Timestamp	0

```
df.duplicated().sum()
```

```
np.int64(0)
```

```
df.tail()
```

Stage 2 – Data Cleaning and Pre-processing

Handle missing values (impute or drop)

Handle duplicates

Treat outliers if required

Check skewness and apply transformations

Convert data types if needed

Feature transformations (date parts, derived fields if required for analysis)

```
df.isnull().sum()
```

Ticket_ID	0
Customer_ID	0
Customer_Name	0
Customer_Email	**0**
Customer_Country	**0**
Customer_Language	0
Channel	0
Query_Category	0
Query_Subcategory	0
Customer_Query_Text	0
Intent_Detected	0
Sentiment_Score	0
Priority_Level	0
AI_Response_Text	0
Human_Escalation_Flag	0
Resolution_Status	0
Resolution_Time	0

Customer_Feedback_Rating	0
Follow_Up_Required	0
Timestamp	0

```
df.duplicated().sum()
```

```
np.int64(0)
```

```
df[['Resolution_Time','Customer_Feedback_Rating','Sentiment_Score']].skew()
```

Resolution_Time	0.005262
Customer_Feedback_Rating	0.041552

Sentiment_Score	-0.0047 44

dtype: float64

```
df['Timestamp'] = pd.to_datetime(df['Timestamp'])

df['Year'] = df['Timestamp'].dt.year
df['Month'] = df['Timestamp'].dt.month
df['Day'] = df['Timestamp'].dt.day

df.to_csv("cleaned_customer_support_data.csv", index=False)
```

Stage 3 – EDA and Visualizations

Univariate Analysis → distribution of single variables (countplot, histogram, boxplot)

Bivariate Analysis → relation between two variables (scatterplot, barplot, correlation heatmap)

Multivariate Analysis → relation among 3+ variables (pairplot, grouped analysis, pivot tables, advanced plots)

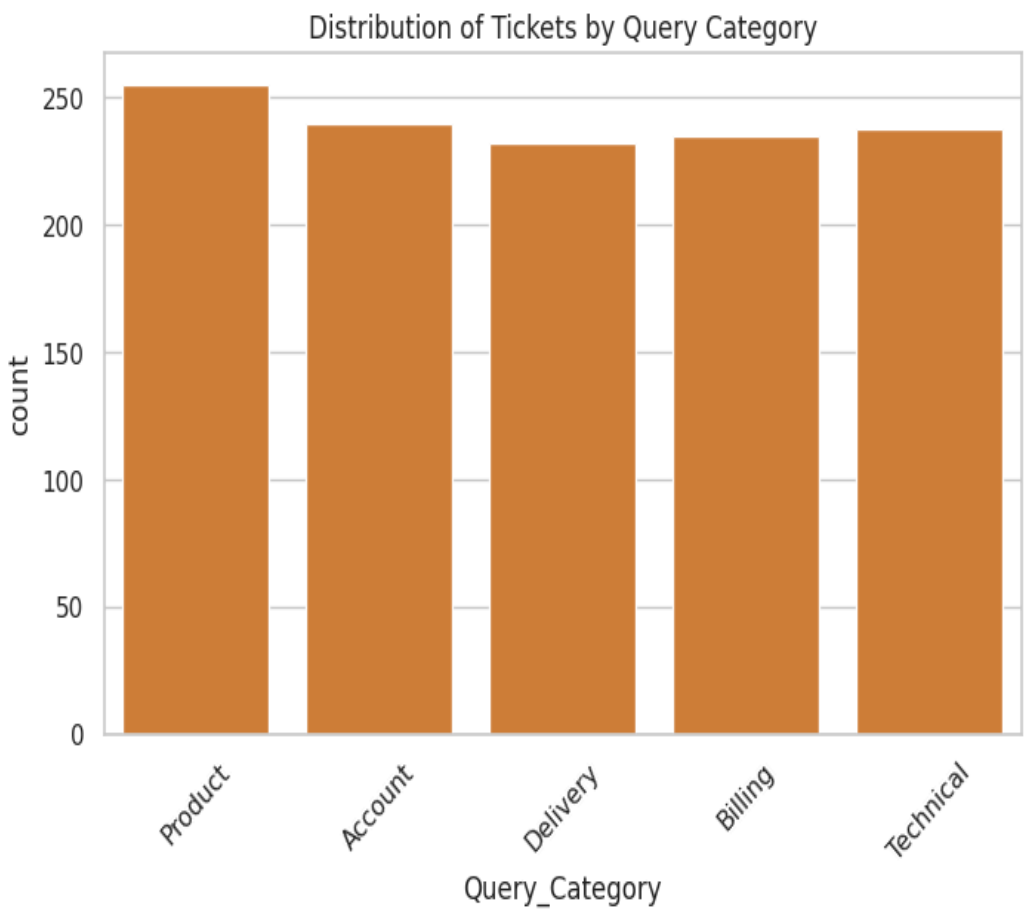
Charts

```
import matplotlib.pyplot as plt
import seaborn as sns

sns.set(style="whitegrid")
# Define theme colors
primary_color = "#E67E22"   # orange
secondary_color = "#8B4513" # brown

from google.colab import files
files.download("cleaned_customer_support_data.csv")

import matplotlib.pyplot as plt
import seaborn as sns
```

```
sns.set(style="whitegrid")

palette = ["#E67E22", "#D35400", "#A04000", "#8B4513"]  # orange-brown theme
```

Distribution of Tickets by Query Category

```
plt.figure(figsize=(8,5))
sns.countplot(x='Query_Category', data=df, color=primary_color)
plt.title("Distribution of Tickets by Query Category")
plt.xticks(rotation=45)
plt.show()
```



Distribution of Tickets by Query Category (Count Plot)

Explain the Chart:

This chart shows the number of customer support tickets grouped by different query categories.

What is it saying?

It indicates which types of issues customers report most frequently.

What features you used?

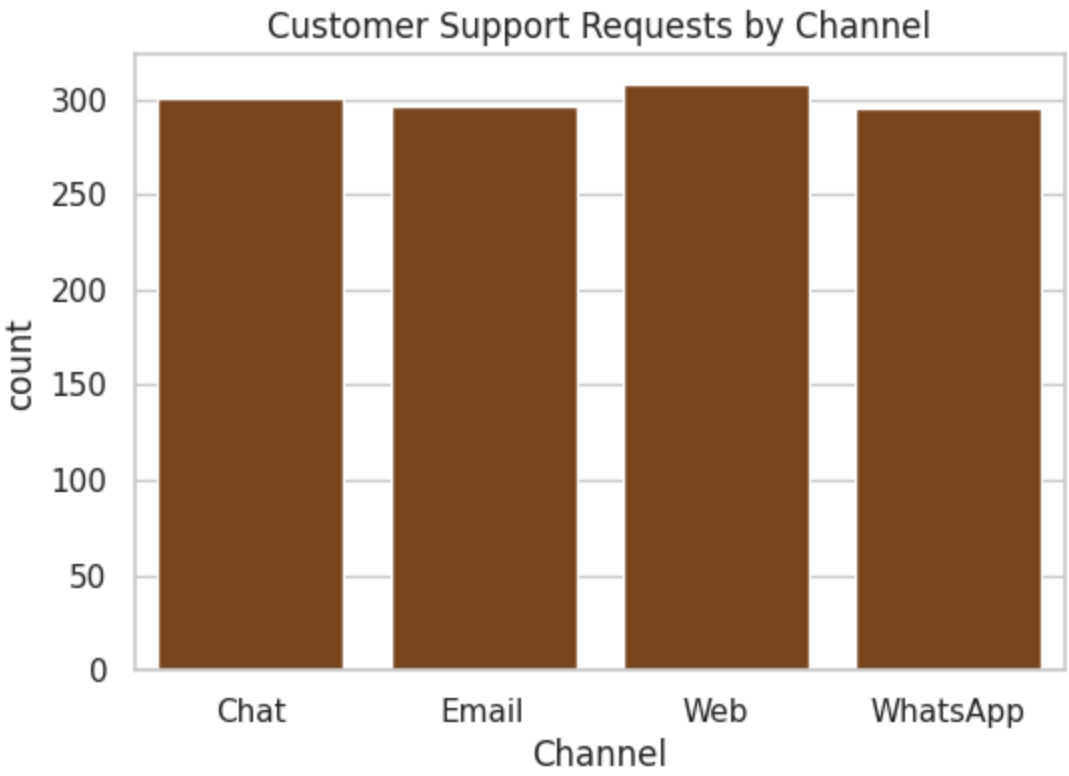
Query_Category.

What does it show?

The chart highlights the most common customer problems and helps identify areas where customer support services may need improvement.

Customer Support Requests by Channel

```
plt.figure(figsize=(6,4))
sns.countplot(x='Channel', data=df, color=secondary_color)
plt.title("Customer Support Requests by Channel")
plt.show()
```



Customer Support Requests by Channel (Count Plot)

Explain the Chart:

This count plot displays the number of customer support requests received through different communication channels.

What is it saying?

It shows which communication channels customers use most frequently when contacting the support system.

What features you used?

Channel.

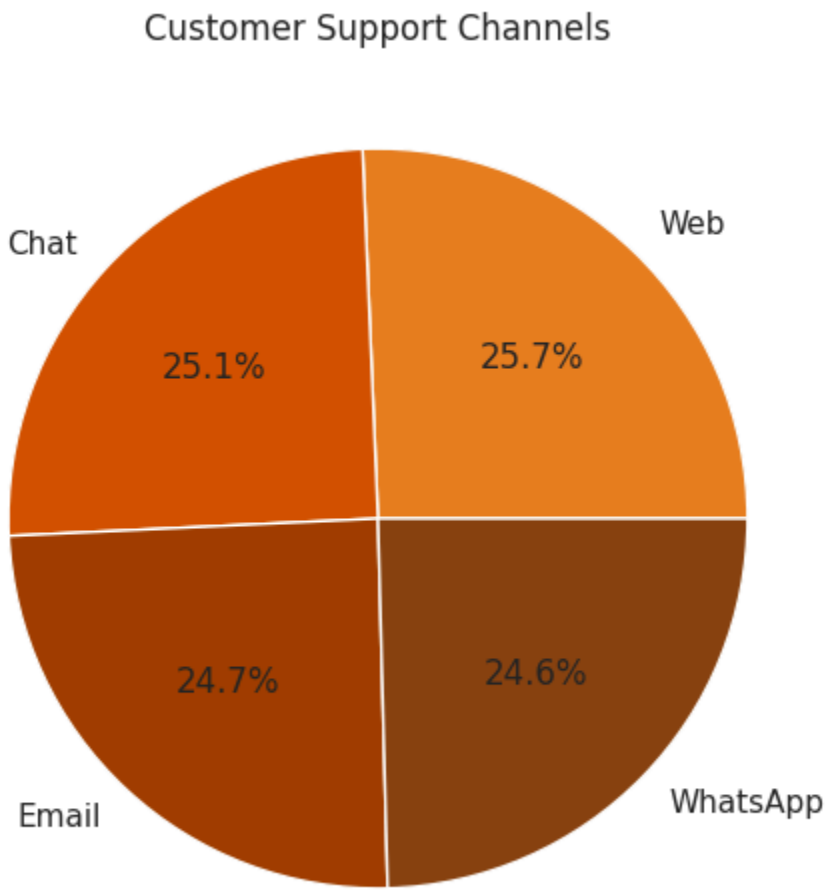
What does it show?

The chart highlights the popularity of different customer support channels such as chat, email, or other platforms. It helps understand customer communication preferences and can guide organizations in improving or prioritizing the most frequently used support channels.

Customer Support Channels

```
plt.figure(figsize=(6,6))

plt.pie(channel_counts, labels=channel_counts.index, autopct='%1.1f%%', colors=palette)

plt.title("Customer Support Channels")
```

```
plt.show()
```



Customer Support Channels (Pie Chart)

Explain the Chart:

This pie chart represents the proportion of customer support requests received through different communication channels.

What is it saying?

\It shows which communication method customers prefer when contacting support.

What features you used?

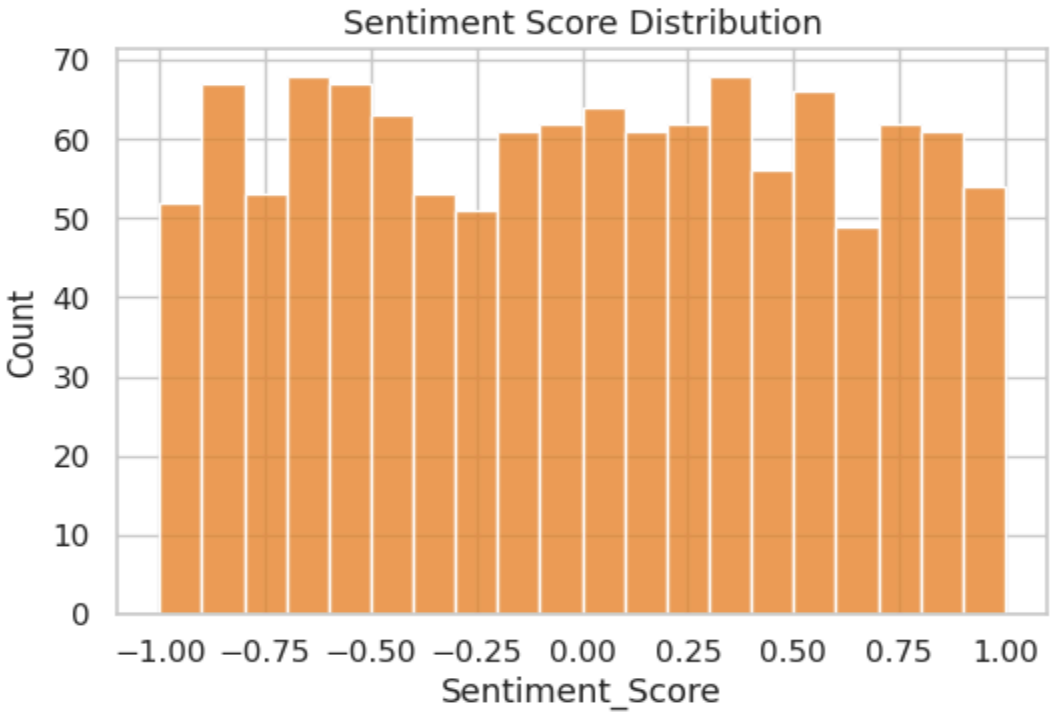
Channel.

What does it show?

The visualization helps understand the popularity of different support channels such as chat or email.

Sentiment Score Distribution

```
plt.figure(figsize=(6,4))
sns.histplot(df['Sentiment_Score'], bins=20, color=primary_color)
plt.title("Sentiment Score Distribution")
plt.show()
```



Sentiment Score Distribution (Histogram)

Explain the Chart.

This histogram shows how sentiment scores are distributed across customer queries.

What is it saying?

It indicates whether customer messages are generally positive, neutral, or negative.

What features you used?

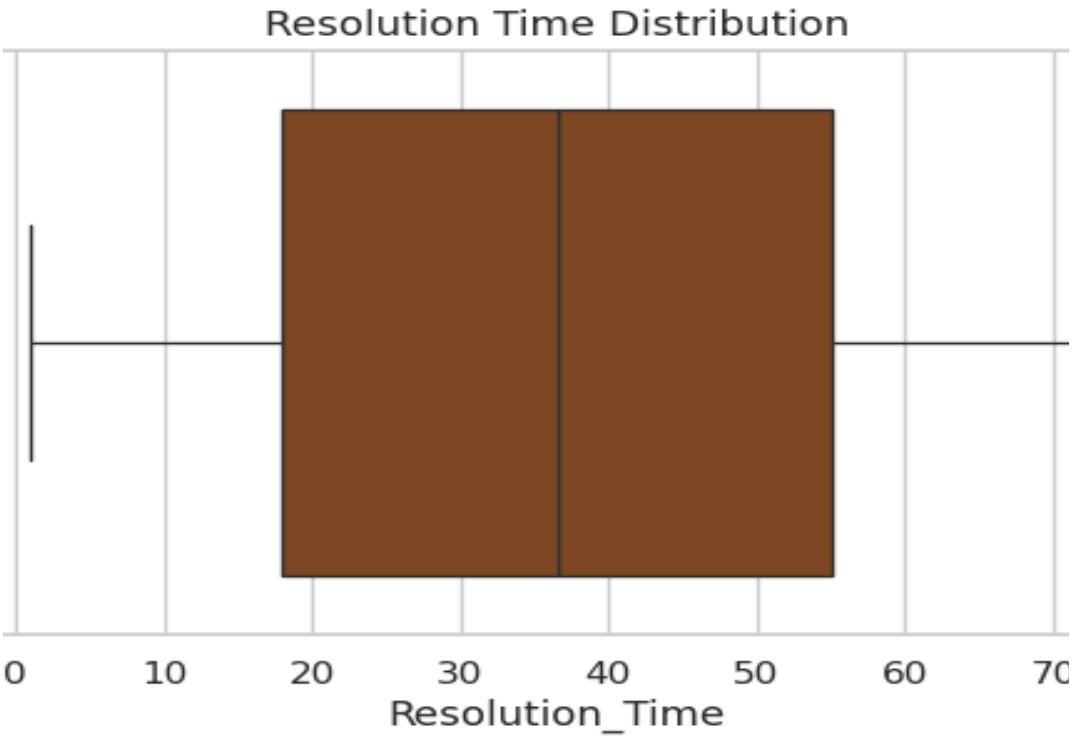
Sentiment_Score.

What does it show?

The chart helps understand the overall emotional tone of customer interactions with the support system.

Resolution Time Distribution

```
plt.figure(figsize=(6,4))
sns.boxplot(x=df['Resolution_Time'], color=secondary_color)
plt.title("Resolution Time Distribution")
plt.show()
```



Resolution Time Distribution (Box Plot)

Explain the Chart.

This box plot illustrates the distribution of resolution times for support tickets.

What is it saying?

It shows the range, median, and spread of time taken to resolve customer issues.

What features you used?

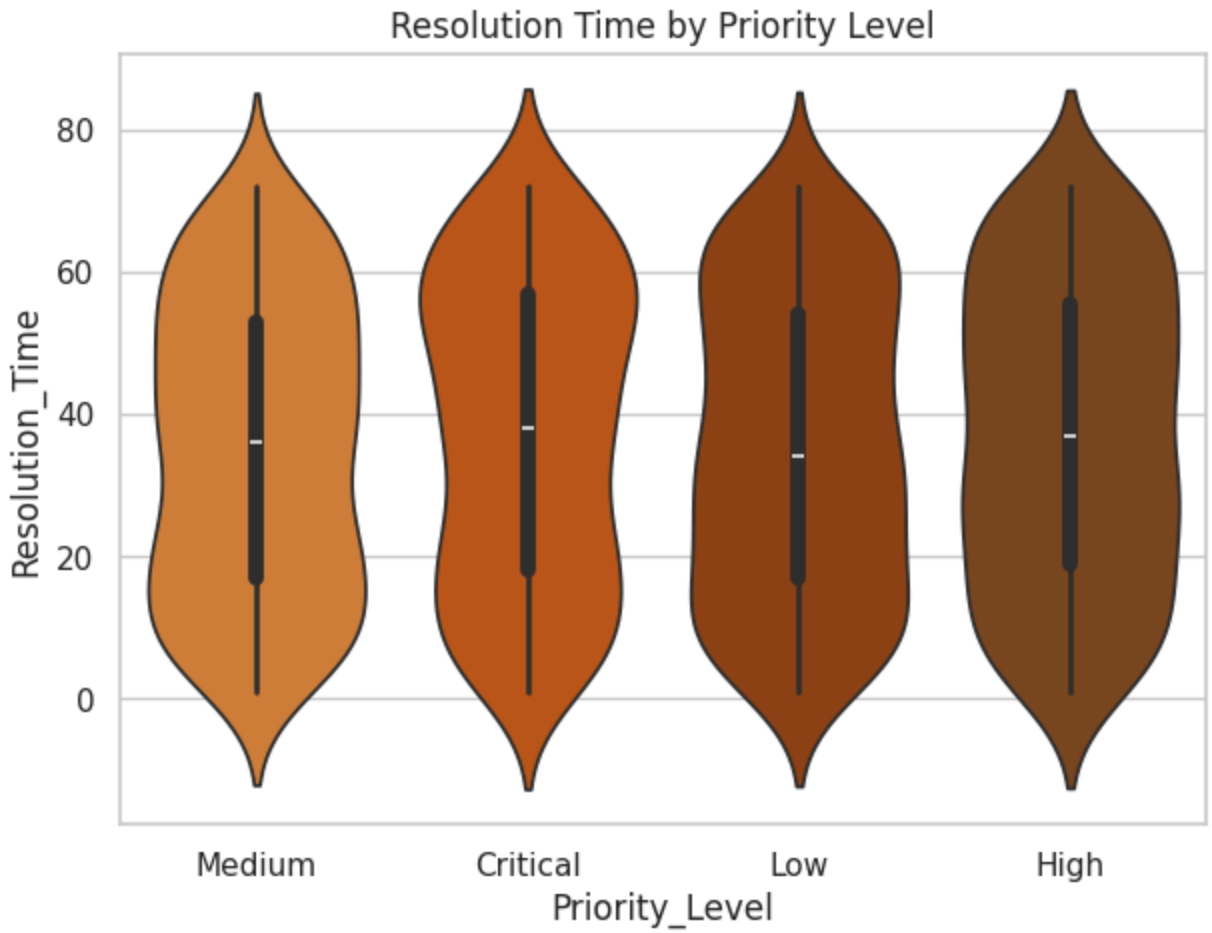
Resolution_Time.

What does it show?

It helps evaluate the efficiency of the support system and identify possible delays.

Resolution Time by Priority Level

```
plt.figure(figsize=(7,5))
sns.violinplot(x='Priority_Level', y='Resolution_Time',
hue='Priority_Level', data=df, palette=palette, legend=False)
plt.title("Resolution Time by Priority Level")
plt.show()
```



t

Resolution Time by Priority Level

Explain the Chart.

This violin plot compares the distribution of resolution times across different priority levels.

What is it saying?

It indicates how quickly issues are resolved depending on their urgency.

What features you used?

Priority_Level and Resolution_Time.

What does it show?

The chart helps determine whether higher-priority tickets are resolved faster than lower-priority ones

Sentiment Score vs Customer Feedback Rating

```
plt.figure(figsize=(6,4))
sns.scatterplot(
    x='Sentiment_Score',
```

```
    y='Customer_Feedback_Rating',
    data=df,
     color="#E67E22",
     alpha=0.6
)
plt.title("Sentiment Score vs Customer Feedback")
plt.show()
```

Sentiment Score vs Customer Feedback
Customer_Feedback_Rating
5.0
4.5
4.0
3.5
3.0
2.5
2.0
1.5
1.0
−1.00
−0.75
−0.50
−0.25
0.00
0.25
0.50
0.75
1.00
Sentiment_Score

Sentiment Score vs Customer Feedback Rating (Scatter Plot)

Explain the Chart:

This scatter plot examines the relationship between sentiment score and customer feedback rating.

What is it saying?

It shows whether positive customer sentiment corresponds to higher satisfaction ratings.

What features you used?

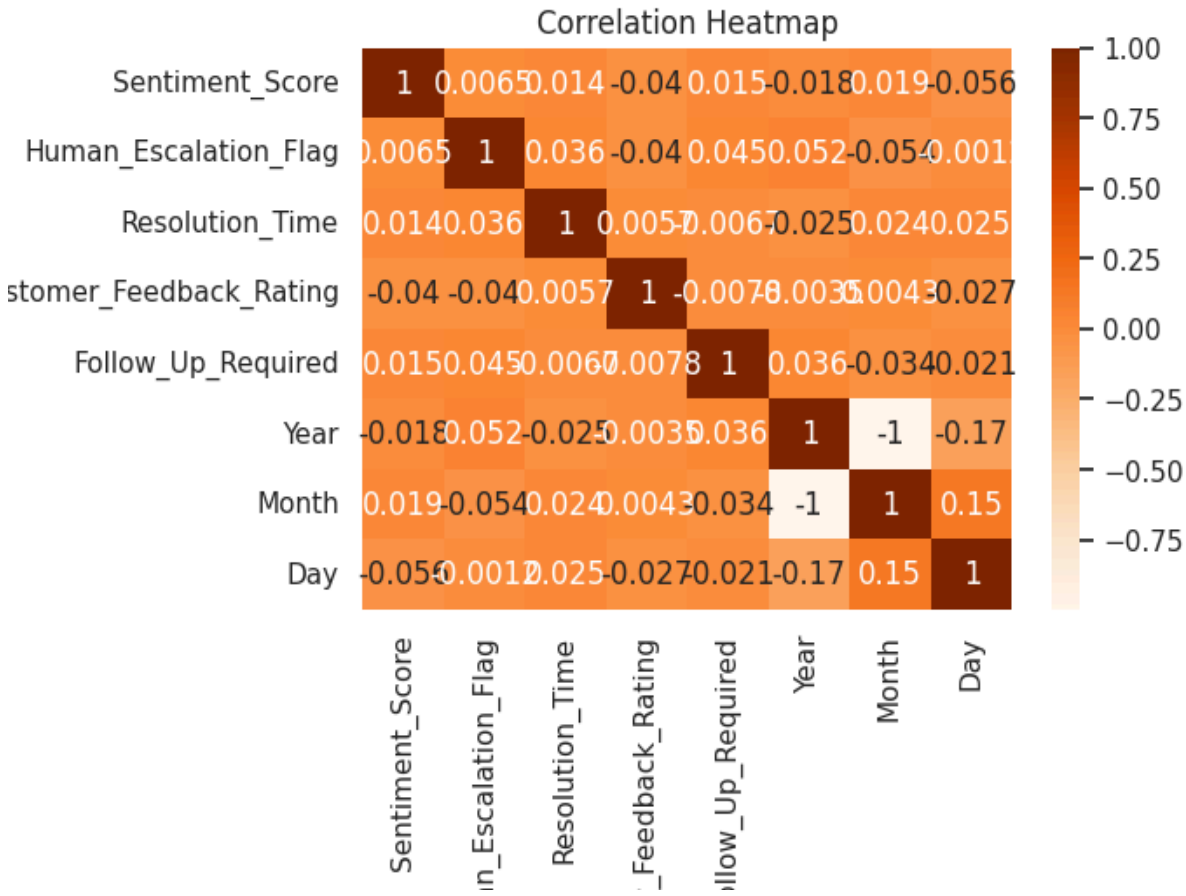
Sentiment_Score and Customer_Feedback_Rating.

What does it show?

The chart helps analyze how customer emotions influence feedback ratings.

Correlation Heatmap

```
plt.figure(figsize=(6,4))
sns.heatmap(df.corr(numeric_only=True), annot=True,
cmap="Oranges")
plt.title("Correlation Heatmap")
plt.show()
```



Correlation Heatmap

Explain the Chart.

The heatmap visualizes correlations between numerical variables in the dataset.

What is it saying?

It indicates the strength of relationships between variables.

What features you used?

Sentiment_Score, Resolution_Time, and Customer_Feedback_Rating.

What does it show?

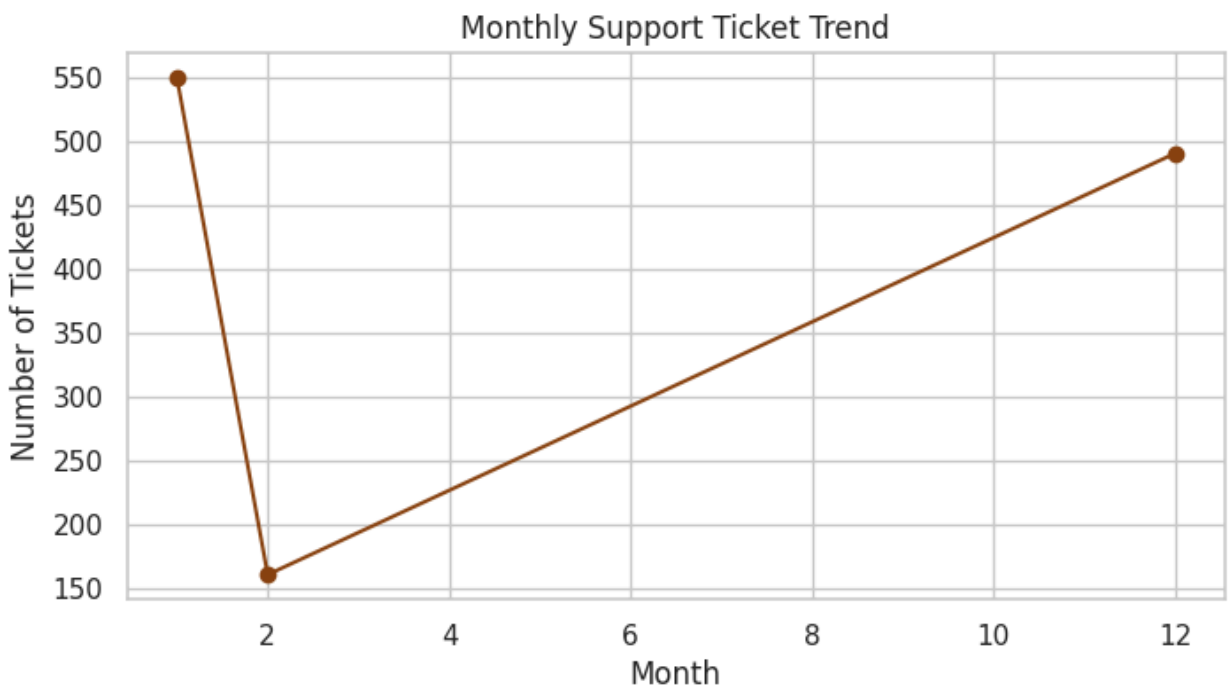
It helps identify whether changes in one variable affect another.

Monthly Support Ticket Trend

```
df['Timestamp'] = pd.to_datetime(df['Timestamp'])
df['Month'] = df['Timestamp'].dt.month

monthly_counts = df['Month'].value_counts().sort_index()
```

Monthly Support Ticket Trend (Line Plot)

```
plt.figure(figsize=(8,4))
plt.plot(monthly_counts.index, monthly_counts.values,
marker='o', color=secondary_color)

plt.title("Monthly Support Ticket Trend")
plt.xlabel("Month")
plt.ylabel("Number of Tickets")
plt.show()
```



Monthly Support Ticket Trend (Line Plot)

Explain the Chart.

This line plot shows how the number of support tickets changes across months.

What is it saying?

It indicates variations or patterns in customer support demand over time.

What features you used?

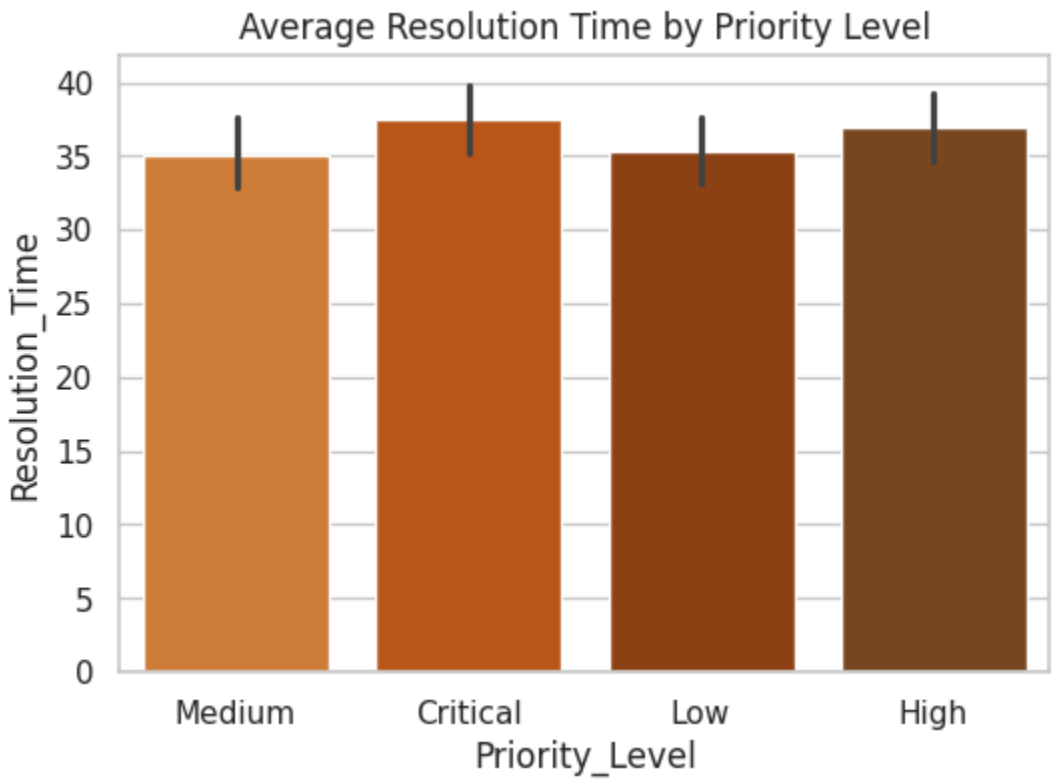
Month (derived from Timestamp).

What does it show?

The chart helps detect seasonal or monthly trends in customer support requests.

Average Resolution Time by Priority Level

```
plt.figure(figsize=(6,4))
sns.barplot(x='Priority_Level', y='Resolution_Time',
data=df, palette=palette)
plt.title("Average Resolution Time by Priority Level")
plt.show()
```



Average Resolution Time by Priority Level (Bar Plot)

Explain the Chart.

This bar chart compares the average resolution time for different ticket priority levels.

What is it saying?

It indicates whether high-priority issues are resolved faster.

What features you used?

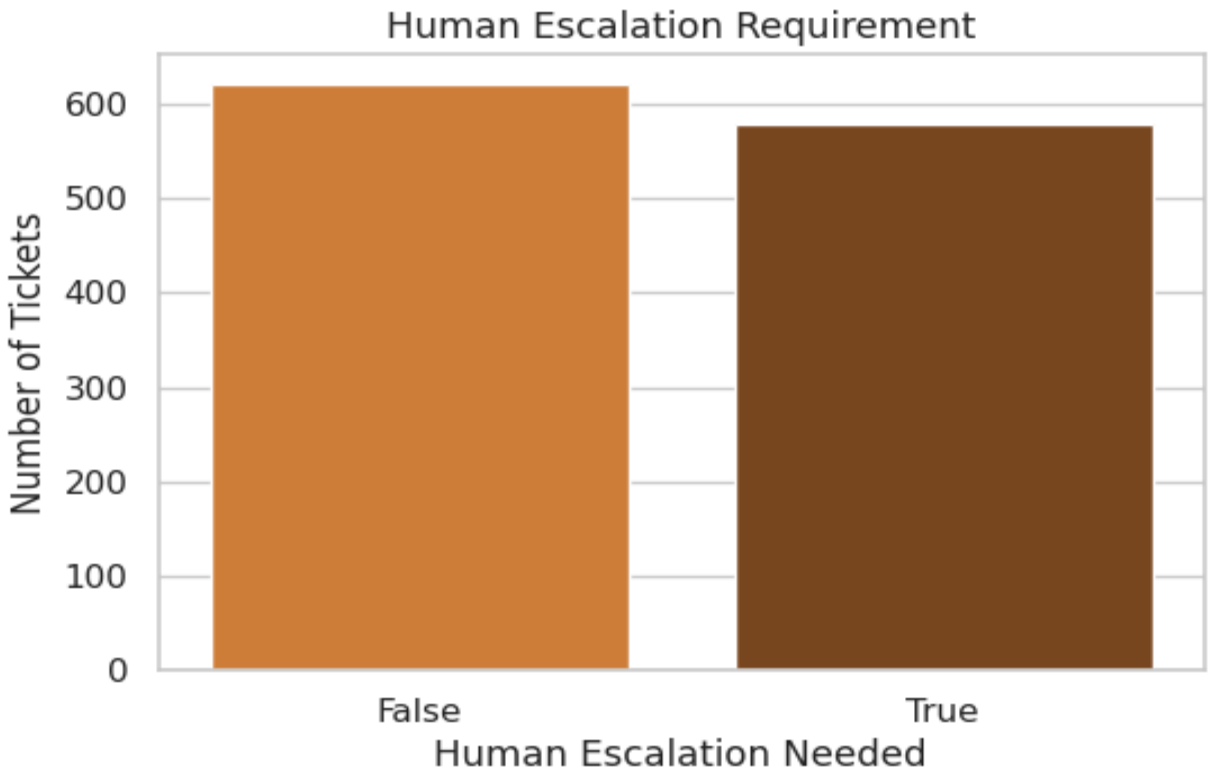
Priority_Level and Resolution_Time.

What does it show?

The chart helps evaluate how efficiently urgent issues are handled.

Human Escalation Requirement

```
plt.figure(figsize=(6,4))
sns.countplot(
    x='Human_Escalation_Flag',
    data=df,
    hue='Human_Escalation_Flag',
    palette=["#E67E22","#8B4513"],
    legend=False
)

plt.title("Human Escalation Requirement")
plt.xlabel("Human Escalation Needed")
plt.ylabel("Number of Tickets")

plt.show()
```



Human Escalation Requirement (Count Plot)

What you can say in your report

Explain the Chart.

This chart shows the number of tickets that required escalation to a human support agent.

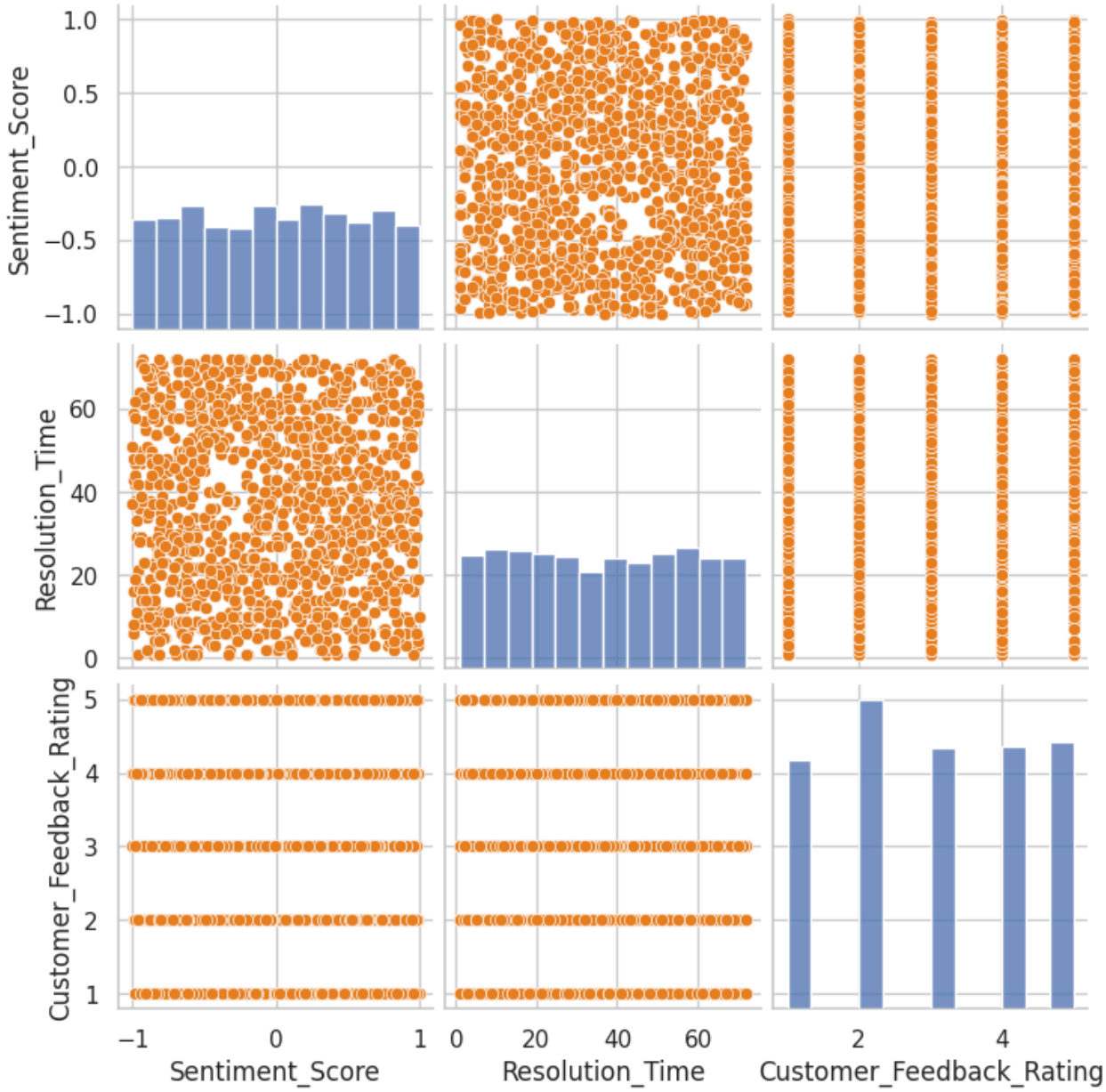
What is it saying?
It indicates how often the AI system needs human assistance to resolve issues.

What features you used?
Human_Escalation_Flag.

What does it show?
The chart helps evaluate the effectiveness of the AI-based customer support system.

Pair Plot – Relationship Between Numerical Features

```
import seaborn as sns
import matplotlib.pyplot as plt
sns.pairplot(

df[['Sentiment_Score','Resolution_Time','Customer_Feedback_Rating']],
    plot_kws={'color':"#E67E22"}
)
plt.show()
```


Sentiment_Score
Resolution_Time
Customer_Feedback_Rating
1.0
0.5
0.0
−0.5
−1.0
60
40
20
0
5
4
3
2
1
−1
0
1
0
20
40
60
2
4

Pair Plot – Relationship Between Numerical Features

Explain the Chart.

The pair plot visualizes the pairwise relationships between the numerical variables in the dataset. It displays scatter plots for each pair of variables and distribution plots along the diagonal.

What is it saying?

It helps identify patterns, trends, or correlations between variables such as sentiment score, resolution time, and customer feedback rating.

What features you used?

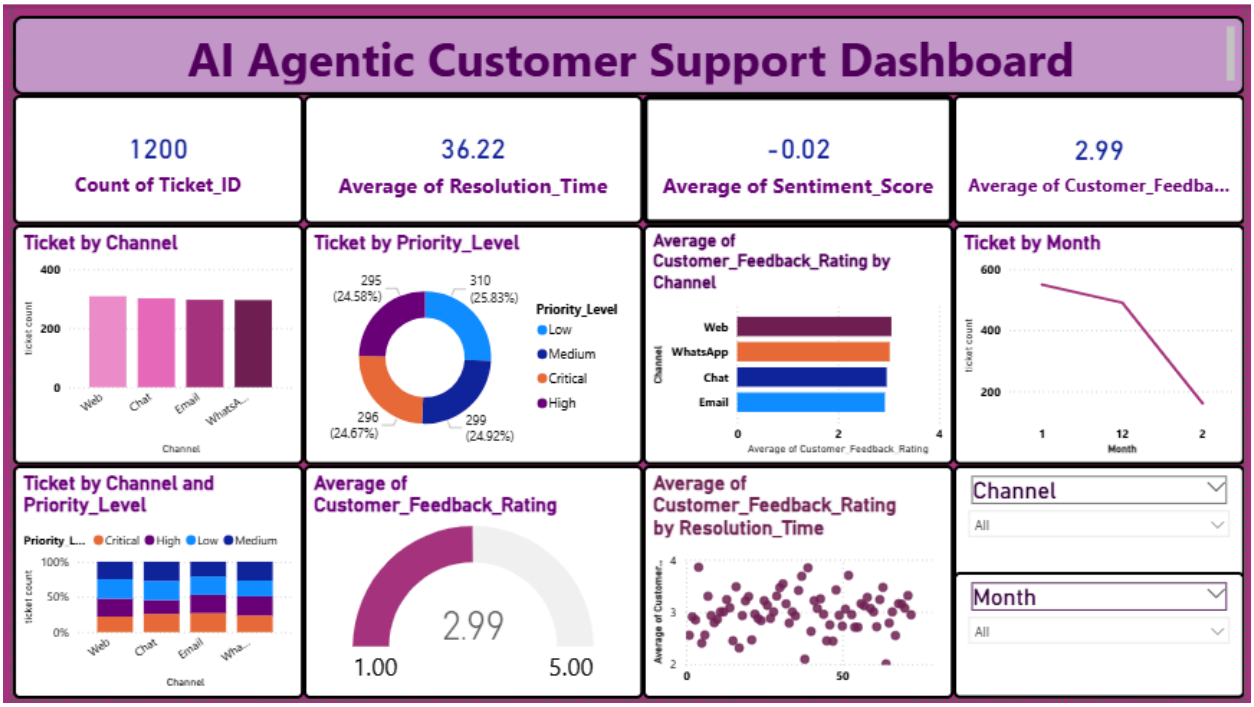
Sentiment_Score, Resolution_Time, and Customer_Feedback_Rating.

What does it show?

The pair plot shows how these variables interact with each other. For example, it helps observe whether higher

sentiment scores are associated with higher customer feedback ratings and whether resolution time has any relationship with customer satisfaction.

Final Dashboard



AI Agentic Customer Support Dashboard – Report

1. Introduction

dashboard is developed to analyze the performance of an AI-based customer support system. It provides insights into ticket volume, resolution efficiency, customer satisfaction, and overall service performance. The

objective is to help stakeholders understand operational effectiveness and improve decision-making.

2. Objectives

- **To analyze the total number of customer support tickets**
- **To evaluate resolution time and efficiency**
- **To To measure customer satisfaction using feedback ratings and sentiment scores**
- **identify trends and patterns in ticket handling**
- **To study the relationship between resolution time and customer satisfaction**

---

3. Key Performance Indicators (KPIs)

The dashboard includes the following KPIs:

- **Total Tickets: Shows the total number of tickets handled (1200)**

- **Average Resolution Time: Indicates the average time taken to resolve issues (36.22)**
- **Average Sentiment Score: Reflects customer sentiment (-0.02, slightly neutral/negative)**
- **Average Feedback Rating: Shows customer satisfaction level (2.99 out of 5)**

---

4. Visualizations and Insights

4.1 Ticket Distribution by Channel

This chart shows how tickets are distributed across different channels such as Web, Chat, Email, and WhatsApp. It helps identify the most frequently used support platforms.

4.2 Ticket Distribution by Priority Level

The donut chart represents the proportion of tickets across different priority levels (Low, Medium, High, Critical). This helps understand workload distribution and urgency levels.

4.3 Average Resolution Time by Priority

This bar chart shows how resolution time varies with priority levels. It helps identify whether high-priority issues are handled efficiently.

4.4 Customer Feedback Rating by Channel

This bar chart displays the average feedback rating for each channel. It helps evaluate which support channel provides better customer experience.

4.5 Ticket Trend by Month

The line chart shows the variation in ticket volume over time. It helps identify patterns such as increase or decrease in customer issues.

4.6 Ticket Distribution by Channel and Priority

This stacked chart shows how different priority levels are distributed across channels, helping in workload and resource analysis.

4.7 Average Customer Feedback Rating (Gauge)

The gauge chart shows the overall customer satisfaction level, providing a quick summary of service quality.

4.8 Scatter Plot: Resolution Time vs Feedback Rating

This advanced visualization shows the relationship between resolution time and customer satisfaction. It helps determine whether longer resolution times negatively impact feedback ratings.

---

5. Key Insights

- **The average feedback rating (2.99) indicates moderate customer satisfaction**
- **Sentiment score is slightly negative, suggesting improvement is needed**
- **Resolution time varies across priorities, indicating potential inefficiencies**
- **Certain channels perform better in terms of customer satisfaction**
- **Scatter analysis suggests a possible relationship between resolution time and feedback**

This dashboard tells the overall story of customer support performance.

Simple Story:

The company handled 1200 tickets

Customers are using different channels (Web, Chat, Email, WhatsApp)

Resolution time is moderate (36.22) → not very fast

Customer satisfaction is average (≈3/5) → needs improvement

Sentiment is slightly negative → customers are not fully happy

“The dashboard shows that while the system handles a significant number of tickets, customer satisfaction is moderate, and improving resolution efficiency can enhance overall service quality.”

BUSINESS RECOMMENDATIONS

1. Improve Resolution Time

Reduce average resolution time
Especially for **high & critical tickets**

2. Focus on High-Usage Channels

Improve support in:

- Web
- Chat

These handle more tickets

3. Improve Customer Satisfaction

Train support agents
Provide faster and better responses

4. Monitor Sentiment

Slight negative sentiment → warning sign
Take feedback seriously**5. Use Data for**

Decisions

Use dashboard regularly to:

- Track performance
- Identify problems early

6. **Future Improvement**

Use AI:

- Auto-response systems
- Predict ticket issue

Final Recommendations

Reduce resolution time, especially for high and critical priority tickets

Improve customer satisfaction by enhancing response quality and support experience

Focus on high-usage channels (Web, Chat) to handle workload efficiently

Monitor sentiment and feedback regularly to identify service issues early

Use data-driven decisions to improve performance and optimize support operations

Conclusion

The dashboard provides a comprehensive view of customer support performance. It highlights key areas

such as resolution efficiency and customer satisfaction. Byanalyzing these metrics, organizations can improve service quality, reduce response time, and enhance customer experience.